Manuel A. Quintero Coronel

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Education

Massachusetts Institute of Technology (MIT), Cambridge, MA, USA 2023 - 2027
PhD in Social and Engineering Systems & Statistics

University of Cambridge, Cambridge, UK 2022 - 2023
Master's in Mathematical Statistics. Concentration: Statistics, Probability, and Optimization

Instituto Tecnológico Autónomo de México (ITAM), Mexico City, Mexico 2015 - 2020
Bachelor of Science in Applied Mathematics

Instituto Tecnológico Autónomo de México (ITAM), Mexico City, Mexico 2015 - 2020
Bachelor of Arts in Economics. Concentration: Econometrics and Economic Theory

Work Experience

PhD Researcher, MIT (IDSS, LIDS, SDSC) Aug 2023 - Current

- Lead author on black-box decomposition methods for evaluating outcome differences across populations; compared to SHAP, ALE, and Sobol approaches; presented at NeurIPS 2024 (oral) and ICLR 2025 (workshop).
- Developed and optimized a Bayesian migration model with MCMC on Supercloud, improving efficiency by 70% and reducing runtime from 60 to 18 days.

Research Assistant, Harvard and ITAM Mar 2021 - Jan 2023

- Co-authored two research projects; led statistical analysis and implemented machine learning models in R and Python.
- Scraped, cleaned, and integrated large datasets from multiple sources; built reproducible data pipelines.
- Applied variable selection and predictive modeling at scale (caret, glmnet), improving robustness and interpretability.

Research Assistant to Fotini Christia (MIT, IDSS Director) Mar 2021 - Aug 2022

- Led statistical analysis for an RCT on gender violence in Egypt, published in *Nature Human Behavior*, using Linear mixed effect models, Difference-in-Differences and other causal inference methods.
- Applied Bayesian methods, variable selection, and cross-validation to ensure robust results and actionable policy insights.

Selected Publications

- Broderick, T., et al. (2025). "Network and Risk Analysis of Surety Bonds." `https://arxiv.org/abs/2511.05691`
- Quintero, M., et al. (2025). Presented at DATA-FM ICLR 2025 and gave an oral presentation at *ATTRIB NeurIPS 2024*. `https://arxiv.org/abs/2504.16864`.
- Christia, F., Larreguy, H., Parker-Magyar, E., et al. (2023). "Empowering Women Facing Gender-Based Violence Amid COVID-19 Through Media Campaigns." *Nature Human Behaviour*. `https://doi.org/10.1038/s41562-023-01665-y`..
- Bandiera, A., Larreguy, H., Parker-Magyar, E., & Quintero, M. (2023). "Can We Shield Citizens Against Misinformation Through Digital Literacy Training and Fact-checks?" Working Paper.

Technical Skills

Programming: Python, R, MATLAB (Advanced); Stata (Intermediate), C++, Java (Basic); **ML/DL Frameworks:** PyTorch, TensorFlow, scikit-learn. **Specialized:** Statistical Machine Learning, Causal Inference. **Tools:** Git, Supercloud HPC. **Collaboration:** Experienced with GitHub, reproducible pipelines, and open-source contributions.

Licenses and Certifications:
Neural Networks and Deep Learning (DeepLearning.AI);
Improving Deep Neural Networks: Hyperparameter Tuning, Regularization, and Optimization (DeepLearning.AI);
Python for Data Science and Machine Learning Bootcamp;
Machine Learning (HarvardX);
High-Dimensional Data Analysis (HarvardX);
Using Python for Research (HarvardX);
Data Science: R Basics (HarvardX).

Selected Honors and Awards

- Fortunato and Catalina Brescia Fellowship, MIT (2025)
- Presidential Graduate Fellowship, MIT (2023-2024).
- FUNED Maestrías Scholarship (Aug 2022).
- Research Award: Honors Thesis in Applied Mathematics (July 2022).
- Research Award: Honors Thesis in Economics (June 2022).